

C++ Interview - Tomorrow Final Revision Guide

This guide is designed for your final-day interview preparation. It contains important interview questions, concise answers, debugging concepts, coding examples, STL, memory management, threading, Linux, and project discussion topics.

1. Output-Based Questions

Q: What is output?

```
class Base {
public:
    virtual void show() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void show() {
        cout << "Derived";
    }
};

int main() {
    Base* ptr = new Derived();
    ptr->show();
}
```

Answer: Output: Derived. Because virtual function enables runtime polymorphism.

Q: What happens here?

```
int x = 10;
int* const p = &x;
```

Answer: Pointer itself cannot change, but value can be modified.

2. Memory Management Questions

Q: Difference between stack and heap?

Answer: Stack memory is automatic and faster. Heap memory is dynamic and manually managed.

Q: Why delete[] for arrays?

```
char* p = new char[100];
delete[] p;
```

Answer: delete[] ensures all array elements are properly released.

3. STL Interview Questions

Q: Difference between vector and list?

Answer: Vector provides fast random access. List provides fast insertion/deletion.

Q: Complexity of map?

Answer: Insertion and search complexity is $O(\log n)$.

4. Coding Questions

Q: Reverse String

```
void reverse(char* str)
{
    int l = 0;
    int r = strlen(str) - 1;

    while(l < r)
    {
        swap(str[l], str[r]);
        l++;
        r--;
    }
}
```

Answer: Uses two-pointer approach.

Q: Palindrome Check

```
bool isPalindrome(string s)
{
    int l = 0;
    int r = s.size() - 1;

    while(l < r)
    {
        if(s[l] != s[r])
            return false;

        l++;
        r--;
    }

    return true;
}
```

Answer: Compare characters from both ends.

5. Debugging Questions

Q: How to debug segmentation fault?

Answer: Use gdb, core dump, backtrace, and check invalid pointer access.

Q: What is valgrind?

Answer: Tool used to detect memory leaks and invalid memory usage.

6. Multithreading

Q: What is mutex?

```
std::mutex mtx;
```

```
mtx.lock();  
// critical section  
mtx.unlock();
```

Answer: Used to protect shared resources between threads.

Q: What is deadlock?

Answer: Two threads waiting indefinitely for each other's resource.

7. Linux Commands

Q: Find all cpp files

```
find . -name "*.cpp"
```

Answer: Searches recursively for all cpp files.

Q: Sort files by size

```
ls -ls
```

Answer: Lists files sorted by size.

8. Project Discussion Questions

Q: Explain your project

Answer: Explain architecture, modules, role, and technologies used.

Q: Biggest bug solved?

Answer: Explain issue, debugging steps, root cause, and final fix.

9. HR Questions

Q: Why should we hire you?

Answer: Highlight debugging skills, C++ experience, ownership, and problem solving.

Q: What are your strengths?

Answer: Quick learning, debugging, teamwork, analytical thinking.

Final Night Revision Checklist

Topic	Important Focus
OOP	virtual, inheritance, polymorphism

Memory	deep copy, leaks, dangling pointer
STL	vector, map, iterators
Coding	strings, arrays, linked list
Debugging	gdb, valgrind
Linux	find, grep, sort, chmod

Interview Day Tips:

1. Do not panic if you don't know an answer immediately.
2. Explain your thinking process.
3. Focus on practical debugging experience.
4. Write clean and readable code.
5. Be confident while explaining your projects.